Ceph 实验手册

一、实验环境介绍

ceph_node1　　192.168.31.11

ceph_node2　　192.168.31.12

ceph_node3　　192.168.31.13

给三台虚拟机分别添加三块硬盘，磁盘大小为 20G

二、给三台虚拟机分别配置 IP 地址，确保能访问外网

#cd　/etc/sysconfig/network-scripts

#vim ifcfg-ens33

YPE=Ethernet

PROXY_METHOD=none

BROWSER_ONLY=no

BOOTPROTO=none

DEFROUTE=yes

NAME=ens33

DEVICE=ens33

ONBOOT=yes

IPADDR=192.168.31.11

PREFIX=24

GATEWAY=192.168.31.1

DNS1=192.168.31.1

#systemctl restart network

其他两台分别设置为 192.168.31.12 和 192.168.31.13，测试三台虚拟机之间能互相通信

且能访问外网

```
[root@ceph_node1 ~]# ping -c 2  www.baidu.com
PING www.a.shifen.com (14.215.177.39) 56(84) bytes of data.
64 bytes from 14.215.177.39 (14.215.177.39): icmp_seq=1 ttl=54 time=24.3 ms
64 bytes from 14.215.177.39 (14.215.177.39): icmp_seq=2 ttl=54 time=24.1 ms

--- www.a.shifen.com ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1004ms
rtt min/avg/max/mdev = 24.153/24.264/24.375/0.111 ms
[root@ceph_node1 ~]# ping -c2  ceph_node2
PING ceph_node2 (192.168.31.12) 56(84) bytes of data.
64 bytes from ceph_node2 (192.168.31.12): icmp_seq=1 ttl=64 time=0.402 ms
64 bytes from ceph_node2 (192.168.31.12): icmp_seq=2 ttl=64 time=1.41 ms

--- ceph_node2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 0.402/0.910/1.418/0.508 ms
[root@ceph_node1 ~]# ping -c2  ceph_node3
PING ceph_node3 (192.168.31.13) 56(84) bytes of data.
64 bytes from ceph_node3 (192.168.31.13): icmp_seq=1 ttl=64 time=0.430 ms
64 bytes from ceph_node3 (192.168.31.13): icmp_seq=2 ttl=64 time=0.553 ms

--- ceph_node3 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 0.430/0.491/0.553/0.065 ms
[root@ceph_node1 ~]#
```

三、设置计算机名

#hostnamectl set-hostname ceph_node1

#hostnamectl set-hostname ceph_node2

#hostnamectl set-hostname ceph_node3

四、关闭所有节点的防火墙和 SELinux

#systemctl disable firewalld

#systemctl stop firewalld

#vim /etc/selinux/config

```
# This file controls the state of SELinux on the system.
# SELINUX= can take one of these three values:
#     enforcing - SELinux security policy is enforced.
#     permissive - SELinux prints warnings instead of enforcing.
#     disabled - No SELinux policy is loaded.
SELINUX=disabled
# SELINUXTYPE= can take one of three values:
#     targeted - Targeted processes are protected,
#     minimum - Modification of targeted policy. Only selected processes are protected.
#     mls - Multi Level Security protection.
SELINUXTYPE=targeted
```

#setenforce 0

五、在所有的节点上配置/etc/hosts 文件

[root@ceph_node1 ~]# cat /etc/hosts

127.0.0.1 localhost localhost.localdomain localhost4 localhost4.localdomain4

::1 localhost localhost.localdomain localhost6 localhost6.localdomain6

192.168.31.11 ceph_node1

192.168.31.12 ceph_node2

192.168.31.13 ceph_node3

六、在 ceph_node1 生成秘钥，并将公钥复制到各个节点，包括自己，建立 ssh 互信

```
[root@ceph_node1 ~]# ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa.
Your public key has been saved in /root/.ssh/id_rsa.pub.
The key fingerprint is:
SHA256:17y2Ni3zCndgg8VZCq+QV5u6FIc62NIscgKdxpZy4/Q root@ceph_node1
The key's randomart image is:
+---[RSA 2048]----+
|          . . .  |
|   o o   . * *   |
|  o @   o + X    |
|   O o = + X     |
|    + E S * *    |
|     + o + o +   |
|          o +..  |
|           +=o.  |
|           .o*.  |
+----[SHA256]-----+
[root@ceph_node1 ~]#
```

#ssh-copy-id -i /root/.ssh/id_rsa.pub ceph_node1

#ssh-copy-id -i /root/.ssh/id_rsa.pub ceph_node2

#ssh-copy-id -i /root/.ssh/id_rsa.pub ceph_node3

#ssh ceph_node2

七、配置时间服务器

1. 在 ceph_node1 上安装 chronyd，其他节点指向该服务器

#vim /etc/chronyd.conf

Allow NTP client access from local network.

allow 192.168.31.0/24 ###允许的网段

Serve time even if not synchronized to a time source.

local stratum 10

systemctl enable chronyd

#systemctl restart chronyd

2. 在 ceph_node2 和 ceph_node3 上指向 ceph_node1

#vim /etc/chrony.conf

server 192.168.31.11　iburst

#server 1.centos.pool.ntp.org iburst

#server 2.centos.pool.ntp.org iburst

#server 3.centos.pool.ntp.org iburst

#systemctl restart chronyd

八、给三个节点配置 yum 源

#wget -O /etc/yum.repos.d/epel.repo http://mirrors.aliyun.com/repo/epel-7.repo

cat << EOF | tee /etc/yum.repos.d/ceph.repo

```
[Ceph]
name=Ceph packages for $basearch
baseurl=http://mirrors.163.com/ceph/rpm-nautilus/el7/\$basearch
enabled=1
gpgcheck=1
type=rpm-md
gpgkey=https://download.ceph.com/keys/release.asc
priority=1

[Ceph-noarch]
name=Ceph noarch packages
baseurl=http://mirrors.163.com/ceph/rpm-nautilus/el7/noarch
enabled=1
gpgcheck=1
type=rpm-md
gpgkey=https://download.ceph.com/keys/release.asc
priority=1

[ceph-source]
name=Ceph source packages
baseurl=http://mirrors.163.com/ceph/rpm-nautilus/el7/SRPMS
enabled=1
gpgcheck=1
```

type=rpm-md
gpgkey=https://download.ceph.com/keys/release.asc
EOF

#yum clean all

#yum list all

部署

一、在 ceph_node1 上安装 ceph-deploy

#yum -y install ceph-deploy

二、在 ceph_node1 上创建一个 cluster 目录，**所有命令在此目录下进行**（文件位置和名字可以随意）

#mkdir /cluster

#cd /cluster

三、在 ceph_node1 上执行部署

ceph-deploy new ceph_node1 ceph_node2 ceph_node3

```
[ceph_node2][      ] Running command: /usr/sbin/ip addr show
[ceph_node2][DEBUG ] IP addresses found: [u'192.168.31.12', u'192.168.122.1']
[ceph_deploy.new][DEBUG ] Resolving host ceph_node2
[ceph_deploy.new][DEBUG ] Monitor ceph_node2 at 192.168.31.12
[ceph_deploy.new][      ] making sure passwordless SSH succeeds
[ceph_node3][DEBUG ] connected to host: ceph_node1
[ceph_node3][      ] Running command: ssh -CT -o BatchMode=yes ceph_node3
[ceph_node3][DEBUG ] connected to host: ceph_node3
[ceph_node3][DEBUG ] detect platform information from remote host
[ceph_node3][DEBUG ] detect machine type
[ceph_node3][DEBUG ] find the location of an executable
[ceph_node3][      ] Running command: /usr/sbin/ip link show
[ceph_node3][      ] Running command: /usr/sbin/ip addr show
[ceph_node3][DEBUG ] IP addresses found: [u'192.168.31.13', u'192.168.122.1']
[ceph_deploy.new][DEBUG ] Resolving host ceph_node3
[ceph_deploy.new][DEBUG ] Monitor ceph_node3 at 192.168.31.13
[ceph_deploy.new][DEBUG ] Monitor initial members are ['ceph_node1', 'ceph_node2', 'ceph_node3']
[ceph_deploy.new][DEBUG ] Monitor addrs are ['192.168.31.11', '192.168.31.12', '192.168.31.13']
[ceph_deploy.new][DEBUG ] Creating a random mon key...
[ceph_deploy.new][DEBUG ] Writing monitor keyring to ceph.mon.keyring...
[ceph_deploy.new][DEBUG ] Writing initial config to ceph.conf...
```

输出没有报错，则表示部署成功，否则会看到报错信息，根据报错信息进行排错

在所有节点上安装 ceph 包

#yum -y install ceph

#ceph -v

ceph version **14.2.15** (afdd217ae5fb1ed3f60e16bd62357ca58cc650e5) nautilus (stable)

 # ceph-deploy mon create-initial　　####生成 mon 角色

#ceph-deploy admin ceph_node1 ceph_node2 ceph_node3

#ceph-deploy mgr create ceph_node1 ceph_node2 ceph_node3　　###提供 web 界面管理 ceph，可选

yum -y install ceph-radosgw

#ceph-deploy rgw create ceph_node1

ceph-deploy mds create ceph_node1 ceph_node2 ceph_node3

初始化 OSD

#ceph-deploy osd create --data /dev/sdb ceph_node1

#ceph-deploy osd create --data /dev/sdc ceph_node1

#ceph-deploy osd create --data /dev/sdd ceph_node1

#ceph-deploy osd create --data /dev/sdb ceph_node2

#ceph-deploy osd create --data /dev/sdc ceph_node2

#ceph-deploy osd create --data /dev/sdd ceph_node2

#ceph-deploy osd create --data /dev/sdb ceph_node3

#ceph-deploy osd create --data /dev/sdc ceph_node3

#ceph-deploy osd create --data /dev/sdd ceph_node3

#ceph -s

```
[root@ceph_node1 ~]# ceph -s
  cluster:
    id:     7cc46e90-a8d7-4e00-8432-8c7705cd1be8
    health: HEALTH_WARN
            Reduced data availability: 1 pg inactive, 1 pg peering
            Degraded data redundancy: 16/561 objects degraded (2.852%), 6 pgs degraded

  services:
    mon: 3 daemons, quorum ceph_node1,ceph_node2,ceph_node3 (age 8m)
    mgr: ceph_node1(active, since 5m), standbys: ceph_node2, ceph_node3
    mds:  3 up:standby
    osd: 9 osds: 9 up (since 4s), 9 in (since 4s); 3 remapped pgs
    rgw: 1 daemon active (ceph_node1)

  task status:

  data:
    pools:   4 pools, 128 pgs
    objects: 187 objects, 1.2 KiB
    usage:   9.1 GiB used, 171 GiB / 180 GiB avail
    pgs:     11.719% pgs not active
             16/561 objects degraded (2.852%)
             21/561 objects misplaced (3.743%)
             102 active+clean
             15  peering
             5   active+recovery_wait+degraded
             4   active+recovering
             1   active+recovery_wait+degraded+remapped
             1   active+recovery_wait+remapped
```

故障排除:

###在 global 字段下添加:

mon clock drift allowed = 2

mon clock drift warn backoff = 30

####向需要同步的 mon 节点推送配置文件

cd /etc/ceph/

ceph-deploy --overwrite-conf config push ceph{01..03}

####重启 mon 服务并验证

systemctl restart ceph-mon.target

[root@host-192-168-0-16 ~]# **ceph pg dump | grep ^22\. | awk '{print $1 "\t" $17}'** ## 22 表示 Pool id ##

dumped all in format plain

22.1 [1,2,3]

22.0 [1,2,3]

22.3 [3,1,2]

22.2 [3,2,0]

22.5 [1,3,2]

22.4 [3,1,2]

22.7 [2,1,3]

22.6 [3,0,2]

管理 ceph

1. 创建存储池

ceph osd pool create mypool1 100

2. 查看已创建存储池信息

#ceph osd pool ls detail

3. 查看副本数及 pg_num
ceph osd pool get mypool1 size
ceph osd pool get mypool1 pg_num
#ceph osd pool set mypool2 pg_num 120
#ceph osd pool set mypool2 pgp_num 120

4. 创建 EC 池，创建 EC 策略
ceph osd erasure-code-profile set ec001 k=3 m=2 crush-failure-domain=osd
#ceph osd pool create mypool2 100 erasure ec001
#ceph osd pool ls detail

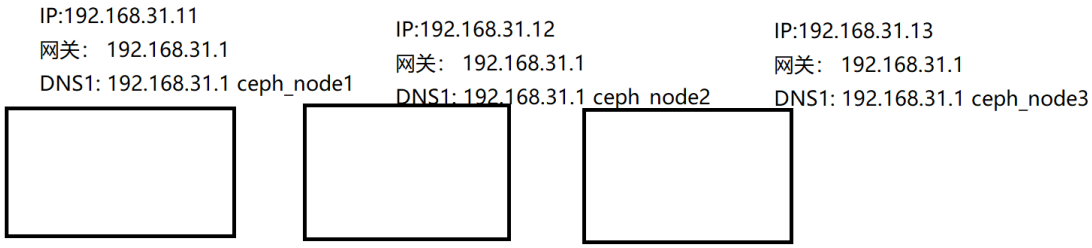
5. 设置 mypool2 为 rgw
ceph osd pool application enable mypool2 rgw
6. 在客户端上传文件到 mypool2，并测试下载
[root@node0 ~]# rados -p mypool2 put passwd.bak /etc/passwd
[root@node0 ~]#rados -p mypool2 get passwd.bak /root/passwd
7. 设置 mypool3 为 rbd，并创建卷，映射给业务服务器使用
ceph osd pool application enable mypool3 rbd
#rbd create mypool3/disk1 --size 1G
[root@node0 ~]# rbd map mypool3/disk1
[root@node0 ~]# ll /dev/rbd0
[root@node0 ~]# mkfs.ext4 /dev/rbd0
[root@node0 ~]#mkdir /data
[root@node0 ~]# blkid
[root@node0 ~]# vim /etc/fstab
UUID="6ed8ed7f-9e2c-4e60-b38c-692d66fe39a9" /data ext4 defaults,_netdev 0 0
[root@node0 ~]# mount -a

[root@node0 ~]# cat /etc/ceph/rbdmap
mypool3/disk1 id=admin,keyring=/etc/ceph/ceph.client.admin.keyring

[root@node0 ~]# cat /etc/ceph/rbdmap
[root@node0 ~]#systemctl enable rbdmap.service
[root@node0 ~]#systemctl restart rbdmap.service

8. 授权
[root@ceph_node1 ~]# ceph auth get-or-create client.alice mon 'allow r' osd 'allow rw pool=mypool2' -o /etc/ceph/ceph.client.alice.keyring
[root@ceph_node1 ~]# scp /etc/ceph/ceph.client.alice.keyring root@node0:/etc/ceph

```
[alice@node0 ~]$ rados    -p mypool2    put group /etc/group --id alice
[alice@node0 ~]$ rados -p mypool2 --id alice ls
```



1. 至少三台服务器/VM，多多益善，可以支持4096节点
2. 2CPU 2G 建议4G
3. 每个节点除OS系统盘外，还需要额外的用于OSD的硬盘，建议每个节点提供三块，20G硬盘
4. 网卡建议两块，一个业务网络，一个存储网络，也可以合并部署，让机器可以访问外网，方便安装软件
5. CentOS7.6，也可以是其他Linux版本

一、修改网络配置，让机器可以访问外网，并且内网互通

二、修改计算机名，添加 hosts 记录

三、关闭防火墙，关闭 SELinux

四、SSH 互信

部署节点 controller --- ceph node

ceph_node1 首节点

五、配置时间服务器，保证各个节点时间一致，如果不一致，系统会警告

专门时间服务器 ---ceph_node1 复用为时间服务器